

TPA ENGINEERING CORPORATION



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Messrs. HYUNDAI WIA CORPORATION
INDIA

Date : Jan. 17, 2024

ESTIMATE

No. TPAS20240117-01

ENQUIRER : NILL

REF. NO. : NILL

MACHINE : TOYODA CNC Production Anguler - Wheel slide Cylindrical Grinder
model GL5Ai-63

PRICE : F.O.B. Busan Port in Korea
- details as per attached sheets -

DELIVERY : PLACE : Busan Port

TIME : 9 months after our receipt of your order note

PAYMENT : T/T (30%:down payment)
(50%:Shipment completed)
(20%: FAC & Warranty Bond Issuance)

VALIDITY : 1 months after the date of this Estimate, or by the date of

We are much pleased in quoting you herein, and thanking for your valued orderin advance.

Yours faithfully,

A handwritten signature in black ink, appearing to be 'JAE HWAN PARK', written over a horizontal line.

JAE HWAN PARK
PRESIDENT

TPA ENGINEERING CORPORATION

[A] TOYODA CNC Angular-Wheelslide Cylindrical Grinder
 Model: 60M GL5Ai-63 (wheel dia. of 610 mm Specification)

* Specially designed and tooled up for
 finish-grinding of OUTER RACE

		Quantity	Unit	Unit Price	Amount
1	<u>Basic Machine</u>				
1.1	Basic machine Model: 60M GL5Ai-63 Wheel diameter ϕ 610 specification	1	set	155,000,000	155,000,000
2.1	Machine bed	1	set	IIBM	IIBM
2.2	One-body type table	1	set	IIBM	IIBM
2.3	Live spindle workhead	1	set	4,200,000	4,200,000
2.4	Footstock, automatic reposition type, manual taper adjustment type	1	set	6,300,000	6,300,000
2.5	Rotary diamond roll type wheel dresser, mounted on the rear of workhead with the inverter	1	set	6,300,000	6,300,000
2.6	Hydraulic pump unit, common with low oil level warning switch	1	set	IIBM	IIBM
2.7	Lubricating and operating oil pump unit common with low oil level warning switch	1	set	IIBM	IIBM
2.8	Coolant supply unit (1,000L) with magnetic separator, paper filter, oil skimmer, English name plates, Spanish PL name plates, high efficiency motor, coolant pump inverter control, caster and coolant cooler	1	set	32,200,000	32,200,000
2.9	JTEKT-make CNC unit	1	set	IIBM	IIBM
2.10	Machine painting (RAL 7035, 1018)	1	set	IIBM	IIBM
2.11	Galvanizing painting	1	set	1,000,000	1,000,000
2.12	Design fee for additional workpieces(total 5kinds)	1	Lot	8,000,000	8,000,000
2.13	Manual overhead open/shut type full cover and totally-closed cover	1	set	7,000,000	7,000,000
2.14	Hydraulic steady rest				not included
2.15	Gentry loader(running speed up specifications)	1	lot	110,000,000	110,000,000
				Sub Total: [1],[2] ₩	330,000,000
3	OPTIONAL ACCESSORIES				
3.1	TOKYO SEIMITSU AMP PULCOM V10A GAGUE PULCOM Σ SERIES Wheel balancing unit, automatic type SB-1000	1	set	45,000,000	45,000,000
3.2	Master bar (for maintenance)	1	set	850,000	850,000
3.3	3-steps signal tower with buzzer	1	set	300,000	300,000
3.4	Cooler for electric control box (RITTAL)	1	set	2,800,000	2,800,000
3.5	Local lighting unit	1	set	200,000	200,000
3.6	Lighting units inside control box	1	set	200,000	200,000
3.7	Electric type jib crane for 0.25ton	1	set	2,000,000	2,000,000
3.8	Air gun & coolant gun	1	set	200,000	200,000
3.9	Mist collector attaching arrangement (excluding mist collector)	1	set	1,030,000	1,030,000

3.10	Terminal box between loader and machine	1 set	500,000	500,000
3.11	Wheel guard auto offset device Front cover automatic correction type, for max. wheel width 125mm	1 set	3,000,000	3,000,000
3.12	Wheel balance unit mounting control			not included
3.13	Balance arbor			not included
3.14	Wheel bearing pump unit (separate type)	1 set	2,000,000	2,000,000
3.15	Dust collector(REVEN)	1 set	5,000,000	5,000,000
3.16	Packing & T/P shipping cost for machine	1 set	5,000,000	5,000,000

Sub Total: [3] ¥¥ 68,080,000

4 Durable tool:

4.1	Grinding wheel flange (by specification 2sets) (5 types)	10 sets	5,500,000	55,000,000
4.2	Feeler & contact for auto-locator (as spare)	1 sets	800,000	800,000
4.3	Feeler & contact for auto-sizer (as spare)	1 sets	1,600,000	1,600,000
4.4	Special type work driving center (L-side only) 5kinds X 2sets	10 sets	1,500,000	15,000,000
4.5	Carbide-tipped work center(R-side only) 1kinds x 2sets	2 sets	300,000	600,000
4.6	Driving plate(for attaching drive center) 5kinds x 2sets	10 sets	150,000	1,500,000
4.7	Loader Jaw (1set = Installed 1set & Spare 1set) for 2 types	4 sets	280,000	1,120,000
4.8	Grinding wheel (by specification 2sets) (5 types)	10 sets	900,000	9,000,000
4.9	Diamond roll, (1set = Installed 1set & Spare 1set) (by specification 2sets)	2 sets	1,500,000	3,000,000
4.10	Paper filter (by specification 2sets)	2 sets	200,000	400,000

Sub Total: [4] ¥¥ 88,020,000

5 Supervisor fee:

5.1	Trial at setup place(INDIA) - included accommodation & domestic transportation fee	1 LOT	34,250,000	34,250,000
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Sub Total: [5] ¥¥ 34,250,000

6 Training & Production support fee:

6.1	Training(operator 3day X 2shift=6M/D & maintenance 3day X 2shift = 6M/I - included accommodation & domestic transportation fee	12 M/D	1,000,000	12,000,000
6.2	EVENT SUPPORT			not included
6.3	Production Support(mechanic 6m/d + electric 6m/d) - included accommodation & domestic transportation fee	12 M/D	1,000,000	12,000,000
6.4	Training translation	12 M/D	300,000	3,600,000
6.5	Manual(usb type 1set, paper type 4set) & Educational Material(paper type	1 lot	2,000,000	2,000,000
6.6	Air Ticket(T/O+training : 2man, Production support : 2man)	4 lot	2,000,000	8,000,000

Sub Total: [6] ¥¥ 37,600,000

7 Spare parts

7.1	Spare parts (3% of Basic machine)			0
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Sub Total: [7] 0

8	<u>Optional items</u>			
8.1	Foundation bar collet type	1	1,800,000	1,800,000
8.2	Work existing confirmation device	1	550,000	550,000
8.3	Tool kit (with torque wrench)	1	300,000	300,000
9	T/P shipping cost (INDIA-> TPA)	1	3,000,000	3,000,000

Sub Total:	[8]	₩	5,650,000
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GRAND TOTAL

₩ 563,600,000